



INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

ASSIGNMENT

Assignment Title: Beginner's Guide to Forensic USB Acquisition, Hash Verification, and Evidence Validation

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Course Title: Basic Computer Skills for Digital Forensics

Course Code: SBT-DF202

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SECTION A: THEORETICAL OVERVIEW

1. DIGITAL INVESTIGATION PROCEDURES & EVIDENCE HANDLING

When investigating digital devices, investigators follow strict steps to make sure any evidence collected can be used in a court of law:

Standard copying (like dragging and dropping files) only copies visible files. A forensic image makes a complete, block-by-block copy of the USB drive. This captures both active files and unallocated space where deleted files, hidden data, or residual system traces live.

Locating physical drives (like USB sticks) that might contain evidence.

Keeping original evidence safe from any changes. Golden Rule: Never work on the original USB drive; always make a clone and perform investigation on the copy.

Making a bit-for-bit forensic image of the USB drive while generating cryptographic hashes (digital fingerprints) to verify that the copy matches the original.

Keeping a detailed paper trail of who handled the USB drive, where it was stored, when it was imaged, and its hash values.

2. RISKS OF USING LOW-LEVEL COMMAND-LINE TOOLS

Using command-line tools like dd or raw disk utilities without a graphical setup introduces real risks for beginners:

- **Risk of Overwriting the Wrong Drive:** Transposing the Input and Output commands can cause the tool to wipe the target evidence USB instead of copying it.
- **No Built-In Write Protection:** Windows automatically tries to read USB drives when plugged in, which can silently alter file modification times or timestamps.
- **Lack of Automatic Verification:** Basic tools copy data without verifying if the copy is identical, unlike dedicated tools (such as FTK Imager) that automatically calculate hashes.

SECTION B: PRACTICAL LAB EXECUTION & EVIDENTIARY ANALYSIS

TASK 1: WORKSPACE CREATION & EVIDENCE PREPARATION

PART A: WORKSPACE PREPARATION

To prepare controlled test evidence, a target directory and evidence files were set up on the USB drive prior to imaging.

A main folder named CIP-B102-Lab4-Evidence was created on the USB drive. Inside, a primary evidence file named ahmed_garba.txt was created, containing student identification details, the creation date, and a confirmation statement. Two additional files were added to test multi-format file validation.

PART B: INITIAL EVIDENCE VERIFICATION & WORKING SPACE

A dedicated folder was created on the host computer's hard drive to safely store the output image file without touching the USB drive:

C:\Users\Ahmed\Forensic_Working_Space

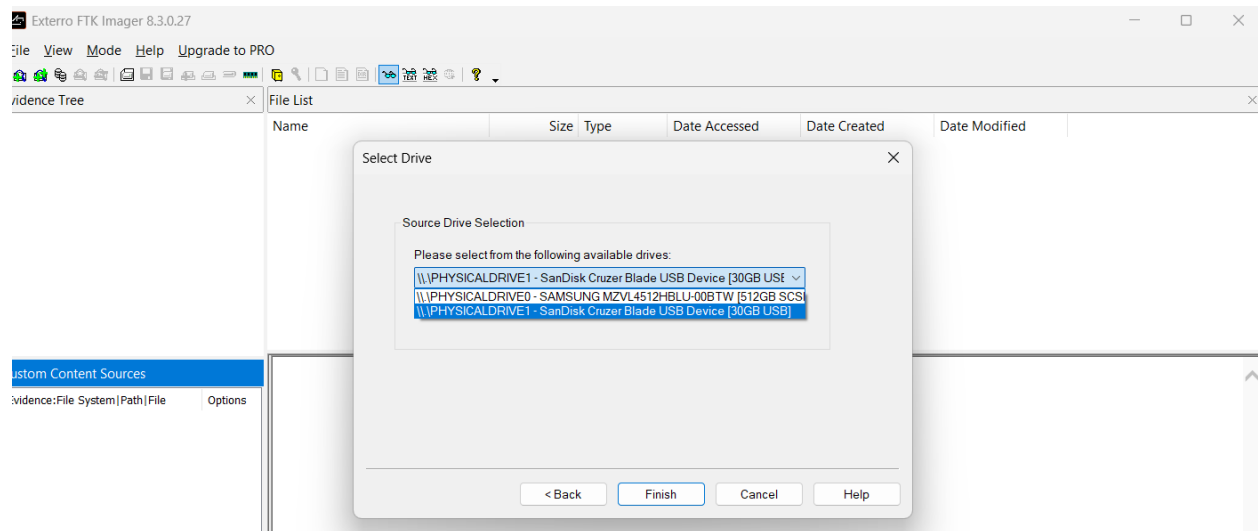
PART C: MINI CHAIN-OF-CUSTODY WORKSHEET

<i>Field</i>	<i>Evidence Entry</i>
<i>Case Title</i>	Digital Forensics USB Acquisition Lab 4
<i>Item ID</i>	Item 001
<i>Evidence Description</i>	SanDisk Cruzer Blade USB Device
<i>Capacity / Sectors</i>	30 GB (7,864,320 total sectors)
<i>Windows Drive Identifier</i>	\\.\PHYSICALDRIVE1
<i>Output Image File</i>	C:\Users\Ahmed\Forensic_Working_Space\usb_image.001
<i>Verification MD5 Hash</i>	7354741c934c2eb44fc25506130f21f0
<i>Verification SHA-1 Hash</i>	db0fbb61b1ebe27277b9fe4ebf55d14d497cb143
<i>Acquired By</i>	Ahmad Zubairu Garba
<i>Acquisition Date</i>	August 29, 2026

TASK 2: FORENSIC ACQUISITION EXECUTION (FTK IMAGER ON WINDOWS)

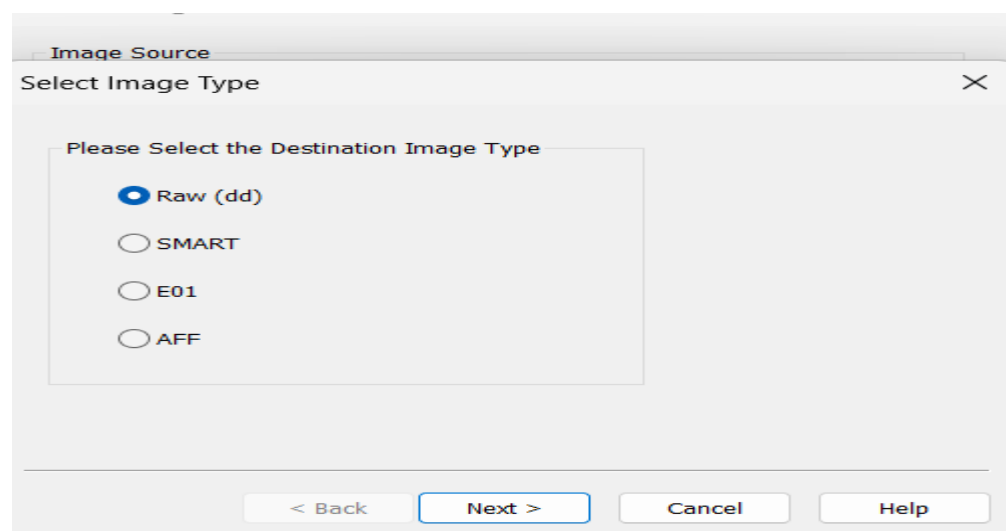
PART A: SELECTING THE SOURCE DRIVE

Exterro FTK Imager v8.3.0.27 was launched as Administrator. Under Create Disk Image, Physical Drive was selected as the source type. The plugged-in USB drive PHYSICALDRIVE1 - SanDisk Cruiser Blade USB Device [30GB USB] was chosen from the drive selection menu.



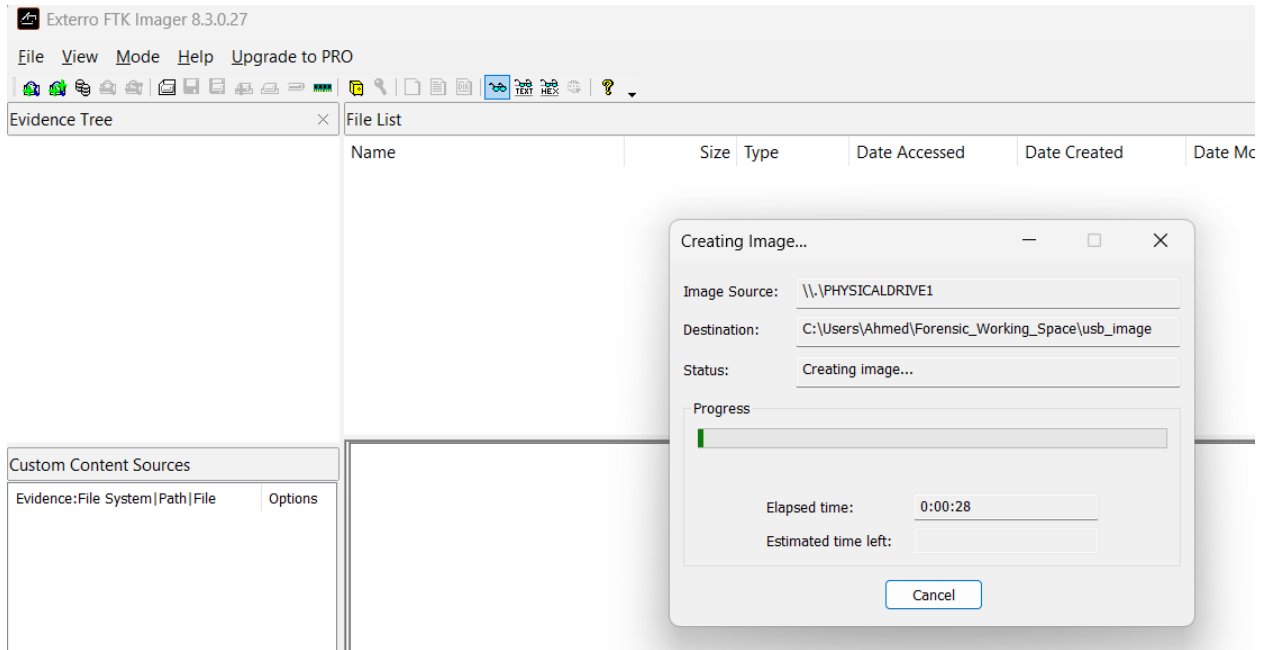
PART B: CONFIGURING IMAGE TYPE AND DESTINATION

The image type was configured to Raw (dd) to ensure an uncompressed, sector-for-sector copy. The destination folder on the computer was specified as C:\Users\Ahmed\Forensic_Working_Space with the filename usb_image. The image fragment size was set to 0 (to keep it as one single file) and automatic image verification was enabled.



PART C: EXECUTING THE IMAGING PROCESS

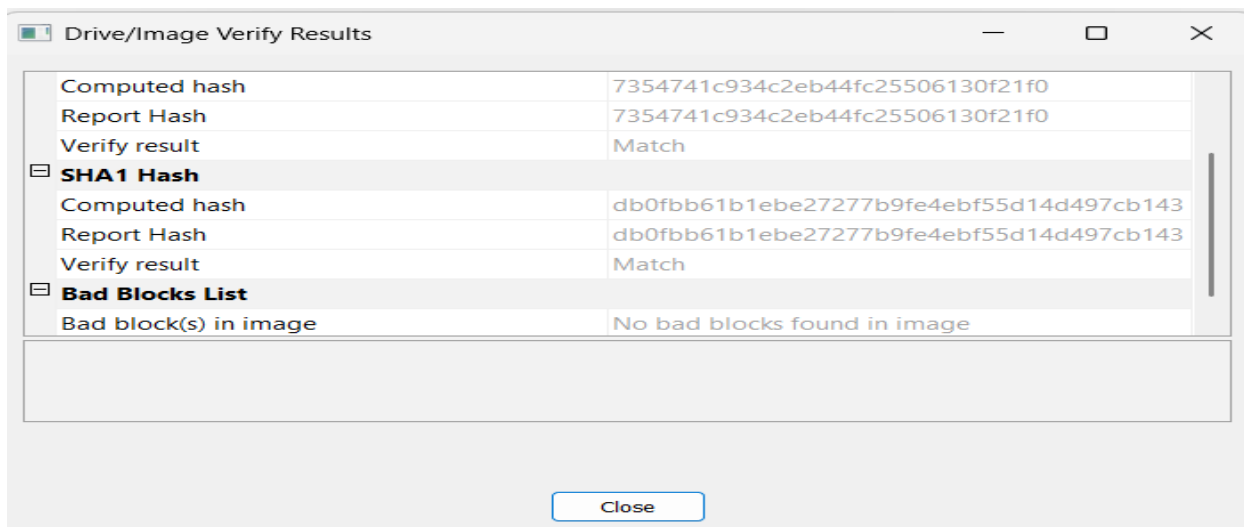
The image creation process was started, reading all 7,864,320 sectors on the USB drive and saving them to the hard drive.



TASK 3: INTEGRITY VERIFICATION & HASH COMPARISON

PART A: HASH COMPUTATION RESULTS

Once imaging finished, FTK Imager automatically generated cryptographic digital fingerprints (MD5 and SHA-1) for both the physical USB drive and the acquired .001 raw image file to verify match integrity.



PART B: INTEGRITY VERIFICATION STATEMENT

The MD5 and SHA-1 hash values calculated directly from the physical USB drive match the hash values of the created raw .001 image file with 100% precision. This mathematical match proves that the image is a perfect clone and no data was modified, corrupted, or altered during acquisition.

TASK 4: EVIDENCE VALIDATION & OUTPUT VERIFICATION

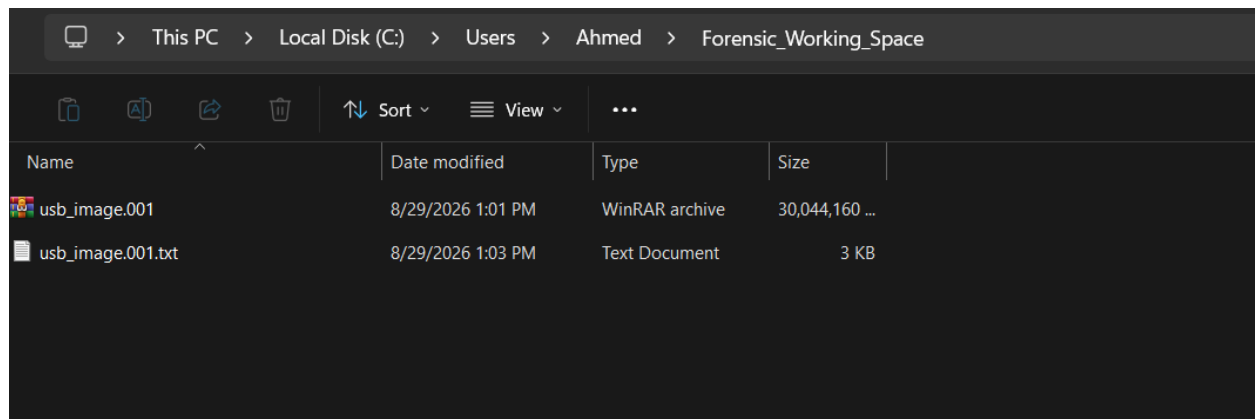
PART A: WINDOWS FILE EXPLORER VERIFICATION

The target folder on the computer was checked in Windows File Explorer to confirm that the image container (usb_image.001) and the verification log (usb_image.001.txt) were created properly:

Directory of C:\Users\Ahmed\Forensic_Working_Space

08/29/2026 01:01 PM 30,044,160,000 usb_image.001

08/29/2026 01:03 PM 3,072 usb_image.001.txt



PART B: VALIDATING FILE CONTENTS IN FTK IMAGER

To confirm that files inside the image were readable without touching the original USB drive, the image file `usb_image.001` was added back into FTK Imager as an Image File evidence item. Opening the directory tree under `CIP-B102-Lab4-Evidence` confirmed that `ahmed_garba.txt`, `sample_document.pdf`, and `sample_image.jpg` were intact and fully accessible.

CONCLUSION

Through completing this lab, I gained practical hands-on experience in the core principles of digital forensics, specifically physical evidence acquisition, integrity verification, and evidence validation.

Key concepts I learned and understood during this practical include:

- **Bit-Stream vs. Standard Copying:** I learned that standard file copying only transfers active files, whereas a forensic raw bit-stream acquisition copies every physical sector (all 7,864,320 sectors in this case). This preserves unallocated space where deleted files and hidden artifacts exist.
- **The Importance of Evidence Isolation:** I understood why forensic investigators must never work on original evidence drives. By creating a raw image (usb_image.001) on the host system, I was able to examine and validate the drive's contents safely without changing any original metadata or timestamps.
- **Cryptographic Hashes as Digital Fingerprints:** I experienced firsthand how cryptographic algorithms like MD5 and SHA-1 work. The matching computed and report hashes 7354741c... for MD5 and db0fbb61... for SHA-1 mathematically proved that the acquired forensic image was a 100% identical clone of the source USB drive.
- **Chain of Custody & Documented Verification:** I learned that proper documentation ranging from recording physical drive paths PHYSICALDRIVE1 to maintaining verification logs usb_image.001.txt is critical for proving evidence integrity and ensuring admissibility in legal proceedings.